

Muhammad Anas Raza

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🎓 Scholar

Summary: Passionate AI/ML researcher with expertise in machine learning, deep learning, computer vision, and data science. Proficient in Python, Pandas, NumPy, Scikit-learn, TensorFlow, and PyTorch. Experienced in exploratory data analysis, data preprocessing, statistical methods, and data visualization (Matplotlib, Seaborn). Skilled in developing and validating ML algorithms on real-world datasets, including image processing and time-series data. Strong communicator with cross-functional collaboration experience. Open-source contributor (Keras).

Education		Skills
Oakland University <i>Ph.D. in Computer Science & Informatics</i> GPA: 3.78 Research Focus: ML in Software Engineering	Rochester, MI May 2022 - Present	Languages & Frameworks: •Python •JavaScript •Keras •TensorFlow •PyTorch •JAX •Streamlit Technology: •Artificial Intelligence •Git •Docker •Machine Learning •Deep Learning •RAG •Web Development •MERN Stack •Large Language Models •Multimodal Learning Soft Skills: •Communication Skills •Public Speaking •Leadership & Mentorship
Oakland University <i>M.S. in Computer Science</i> GPA: 3.76 Courses: •Visual Computing •Object Oriented Analysis •Advanced Data Structures & Algorithms	Rochester, MI May 2022 - Apr 2024	

WORK EXPERIENCE

Oakland University — Graduate Research & Teaching Assistant	MAY 2022 - PRESENT
• <i>Software Engineering & LLM Agent Research:</i> Investigate automated software engineering techniques, including the Graph Neural Networks for fault localization in machine learning systems {P1} and the development of multimodal automated machine learning (AutoML) pipelines. Automated Software engineering using LLM agents. • <i>Computer Vision Research and Development:</i> Conduct research on deepfake detection using Vision Transformers, MLP-Mixer, gMLP, FNet, and Convolutional Neural Networks. Perform experiments in both vision-only {P2} and multimodal {P3}, neuro-symbolic {P4} settings on large-scale datasets including DFDC (124K videos) and FakeAVCeleb (21K videos). • <i>Deep Forgery Detector:</i> Designed and developed in-house Deep Forgery Detector, a modular framework for benchmarking and comparing video deepfake detection algorithms, with extensible architecture enabling rapid integration of new research models. • <i>Forensic Examiner</i> Developed a flexible framework for evaluating audio spoofing detection algorithms, built using Python, PyTorch, TensorFlow, and Streamlit for interactive experimentation and analysis. • <i>Teaching Assistant</i> : Assisted in courses on <i>Machine Learning</i> and <i>Software Testing</i>, including grading assignments, supporting laboratory sessions, and mentoring students on programming and machine learning projects.	

SenseDigital — AI Software Engineer	DECEMBER 2020 - JULY 2021
• <i>IoT Data Pipeline:</i> Designed and implemented automated data ingestion pipelines for time-series data from building automation systems using BACnet, enabling large-scale monitoring of HVAC and building efficiency metrics. • <i>Edge AI Infrastructure:</i> Developed and deployed edge-based data processing and monitoring solutions on Nvidia Jetson Nano devices for real-time building analytics. • <i>Data Visualization & Deployment:</i> Built and containerized data visualization and monitoring dashboards for building efficiency products using InfluxDB and Grafana, and deployed them on Linux-based edge devices. • <i>Tech Stack:</i> Python, Linux (Arch), Nvidia Jetson Nano, Volttron, Docker, InfluxDB, Grafana, Bash.	

TECHNICAL PROJECTS

K3IM: Image Models in Keras — Open Source	DEC 2023 - FEB 2024
• Developed K3IM, an open-source Keras 3 library implementing 50+ state-of-the-art architectures (ViT, Swin, ConvMixer) with native support for 1D, 2D, and 3D/spatiotemporal data. Engineered for multi-backend compatibility, the project enables seamless execution across TensorFlow, PyTorch, and JAX, integrating advanced research features like Class-Attention and Focal Modulation to provide a flexible, high-performance framework for modern machine learning research. Link: https://github.com/anas-rz/k3im	
K3 Addons: Addons for Keras 3 — Open Source	DEC 2023 - MARCH 2024
• Developed K3-Addons, an open-source suite of specialized layers, losses, and activations that extends the Keras 3 ecosystem with multi-backend support (JAX, PyTorch, TensorFlow). I engineered high-performance implementations of modern components, including Attention mechanisms (MobileViT, CBAM, ECA), Adaptive Pooling layers parity-matched with PyTorch, and advanced loss functions like GIoU and Sigmoid Focal Cross-Entropy. Link: https://github.com/anas-rz/k3-addons	
K3NODE: Multibackend Graph Neural Networks — Open Source	JAN 2024 - FEB 2024
• Developed K3Node, a high-performance library for Graph Neural Networks (GNNs) built on Keras 3, porting and optimizing core components from Spektral, PyTorch Geometric, and StellarGraph. I implemented a modular suite of GNN layers (GCN, GAT, GraphSAGE, GIN) and advanced aggregation schemes like PowerMean and Softmax pooling, ensuring cross-framework compatibility. Link: https://github.com/anas-rz/k3-node	

Keras 3 Development — Open Source	JUL 2023 - SEP 2023
• Contributed 20+ merged pull requests to the Keras 3 codebase, developing framework-agnostic implementations of complex architectures like DeepLabV3Plus, Vision Transformers, and SimSiam. Optimized core library operations, including patch extraction ops, and authored comprehensive examples for Reinforcement Learning and Computer Vision, ensuring robust performance across JAX, PyTorch, and TensorFlow backends. Accepted PRs: {Contributions}	